

MARKETPLACE INTELLIGENCE PLATFORM

Version 1.0 / Portfolio Hero

Business Intelligence • Customer Analytics • Network Science • Decision Intelligence



Project Scope

A complete marketplace analytics system: data atlas, economic structure, executive intelligence, customer segmentation, seller risk, product-category analysis, network fragility and intervention ROI.

Demonstrates: Power BI thinking • SQL-ready model • Python analytics • dashboards • decision recommendat

Portfolio Signal

Unlike a standard dashboard, this platform moves from descriptive BI to prescriptive decision intelligence: what happened, why it matters, where the risk is, and what action should be prioritized.

Target roles: Data Analyst • Business Analyst • Power BI Analyst • Analytics Engineer Jr

EXECUTIVE INTELLIGENCE CENTER

Revenue • Orders • Concentration • Marketplace Performance

Dashboard 1 / Marketplace BI

REVENUE

\$13.59M

ORDERS

99,441

CUSTOMERS

96,096

SELLERS

3,095

AOV

\$138

LATE RATE

6.8%

REVIEW

4.09

Revenue by Category



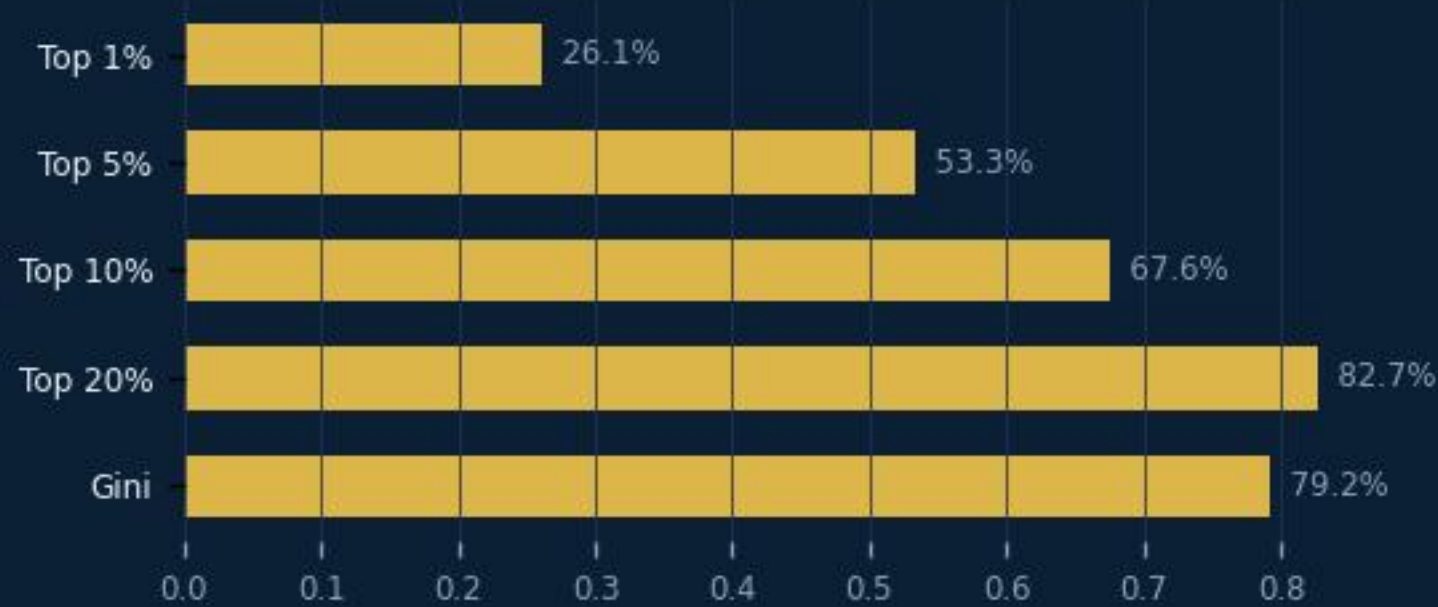
Monthly Revenue Trend



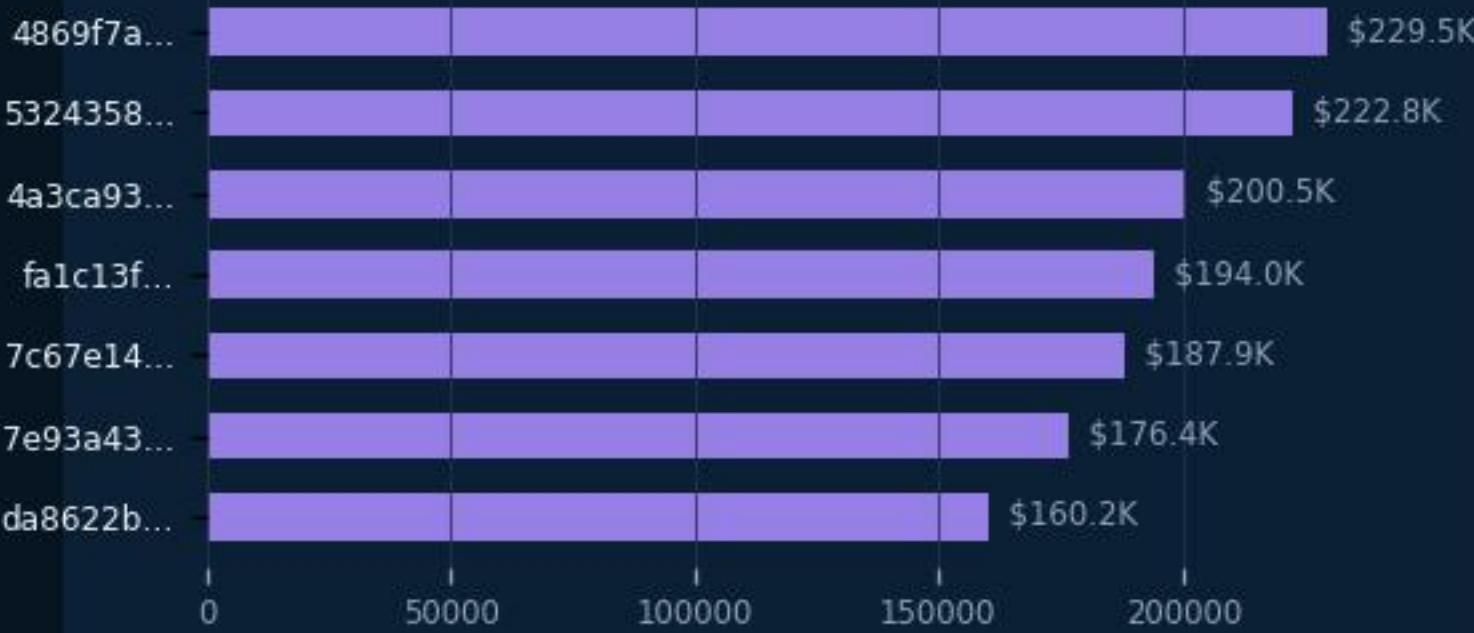
Revenue by Customer State



Economic Concentration



Top Sellers



Executive Reading

Core message

The marketplace generated \$13.59M across 99,441 orders. Top seller concentration is material: the top 1% of sellers explains 26.1% of revenue.

Business layer demonstrated: KPI design, revenue intelligence, concentration analysis, executive reporting.

CUSTOMER INTELLIGENCE ENGINE

RFM Segmentation • Retention • Customer Value • Cohort Behavior

SEGMENTS

6

CUSTOMERS

96,096

SEGMENT REVENUE

\$13.59M

TOP SEGMENT

High-value one-time buyers

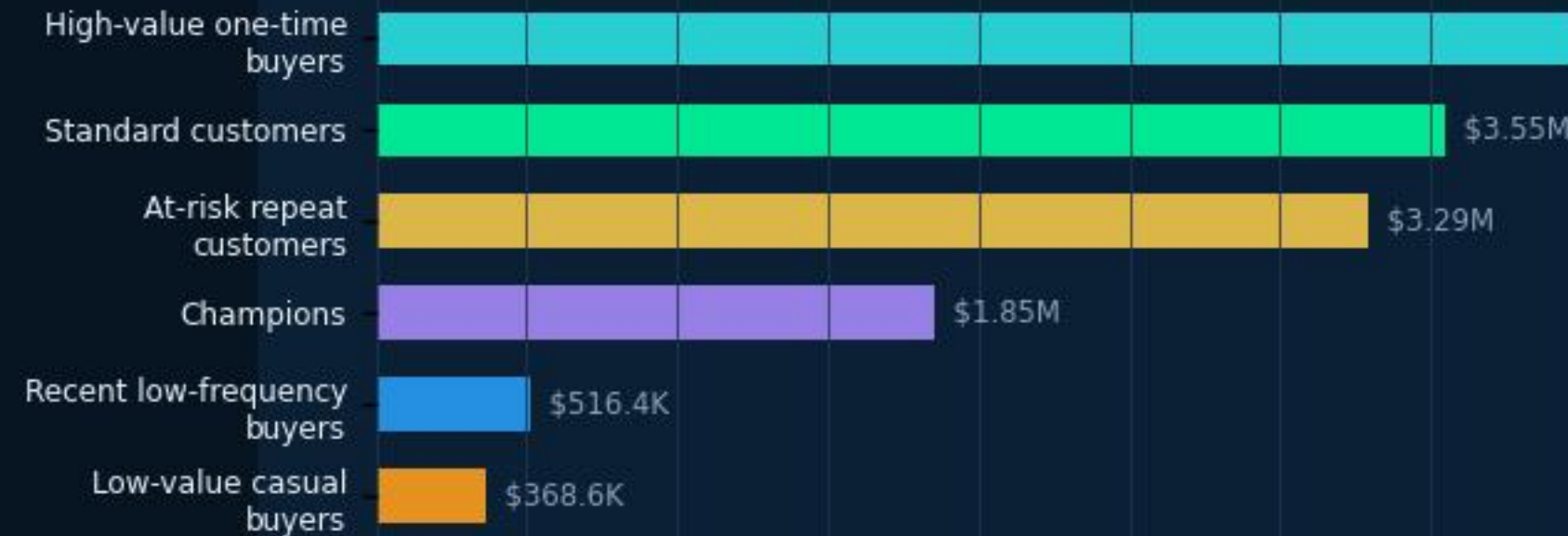
REPEAT RATE

3.7%

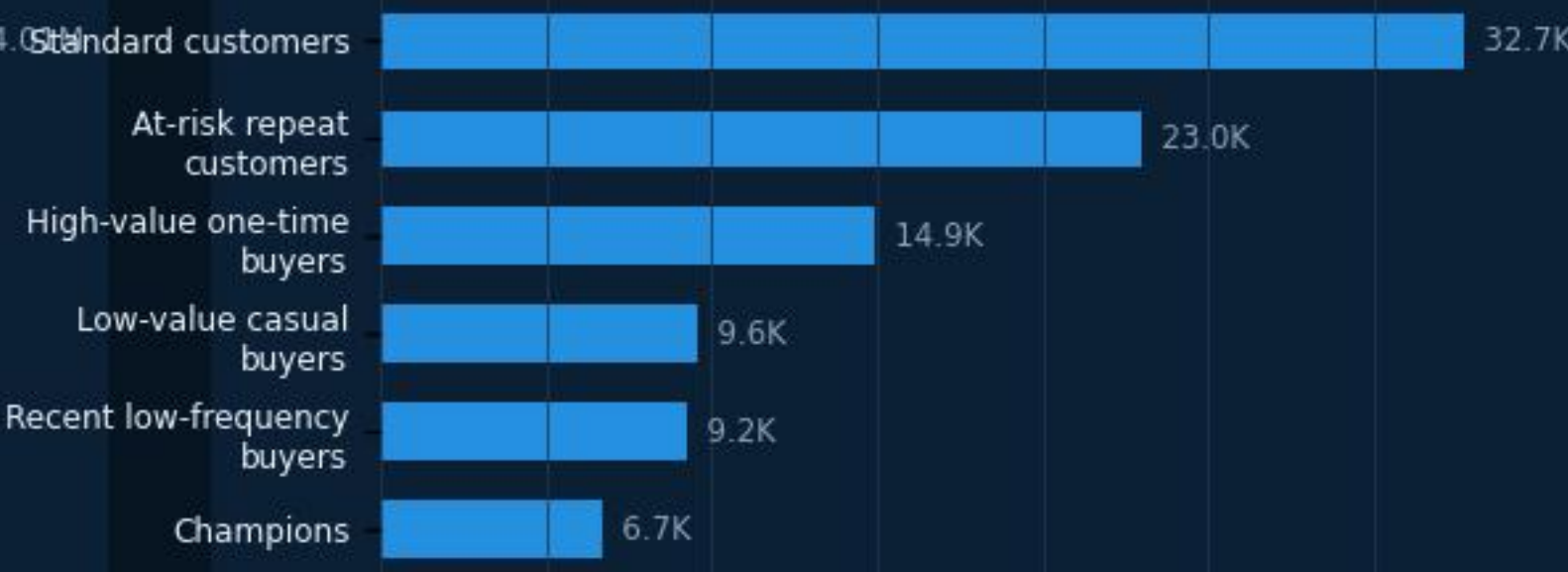
M1 RETENTION

4.4%

Revenue by Customer Segment



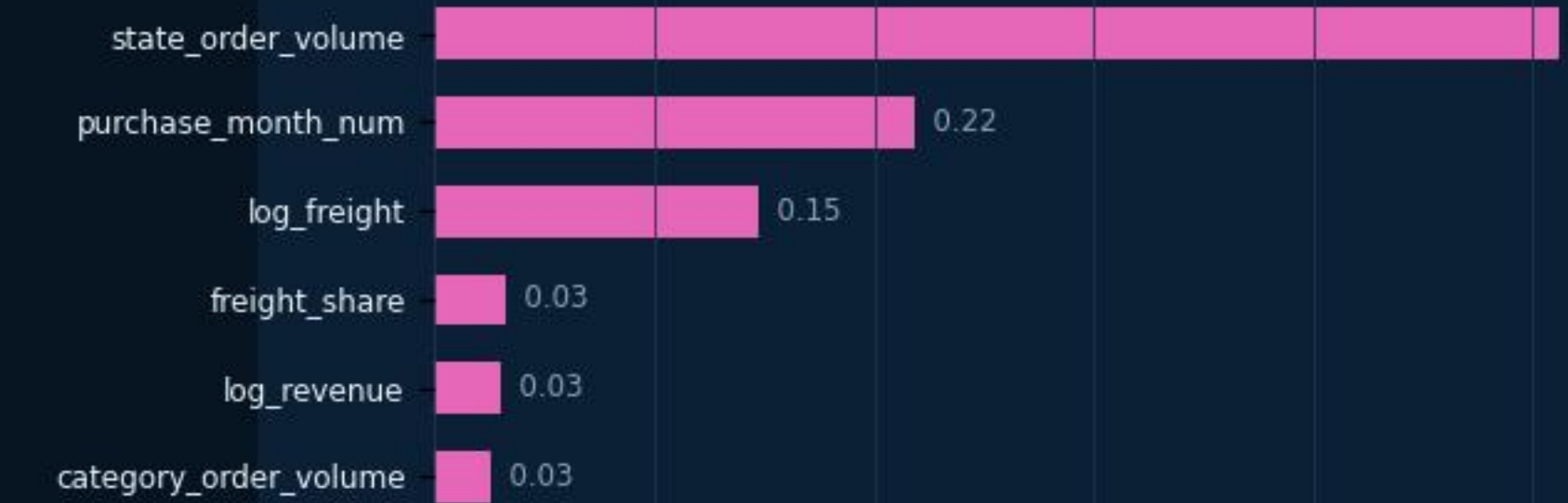
Customers by Segment



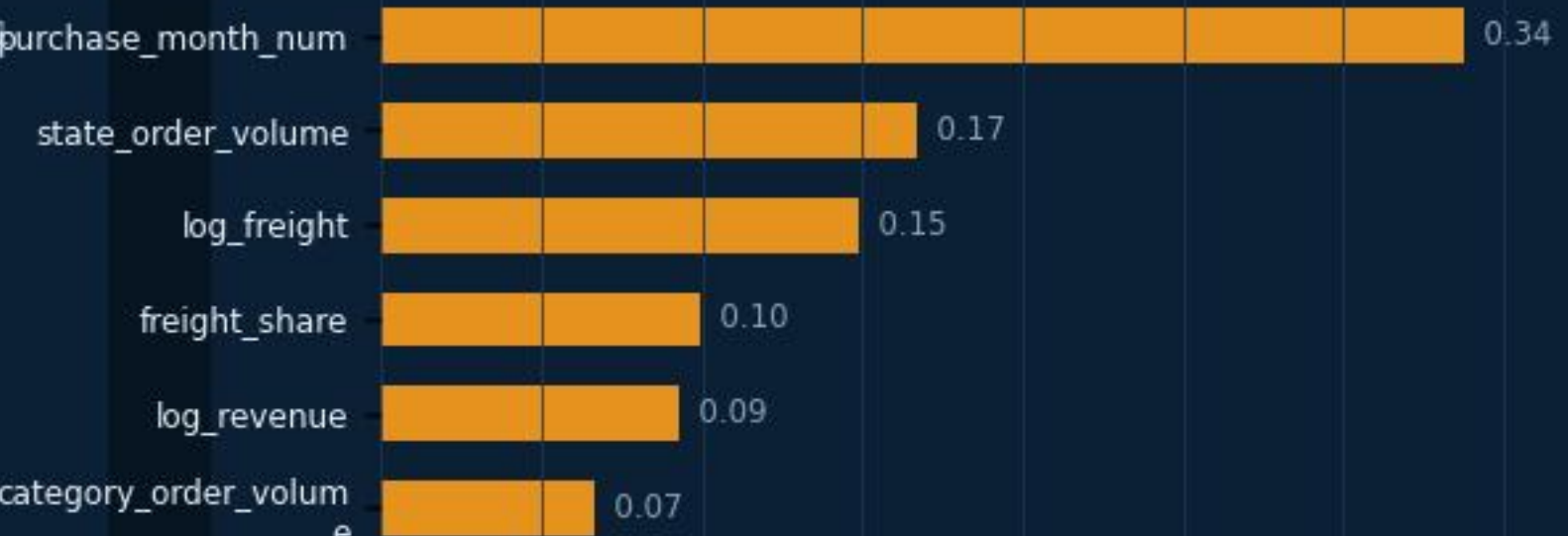
Average Retention Curve



Delivery Time Model Drivers



Late Delivery Drivers



Customer Decision Reading

Core message

High-value one-time buyers is the largest value pool with \$4.01M in revenue. Customer intelligence converts the marketplace from a transaction table into retention, value and targeting logic.

Analytics layer demonstrated: RFM segmentation, retention proxy, cohort curve, customer value analysis.

SELLER INTELLIGENCE & RISK CENTER

Seller Segmentation • Review Risk • Operational Quality • Intervention Targets

SELLER SEGMENTS

5

ANALYZED SELLERS

1,271

TOP RISK SELLER

712e6ed8...

TOP CATEGORY

health_beauty

RISK DRIVER

delivery_delay_da

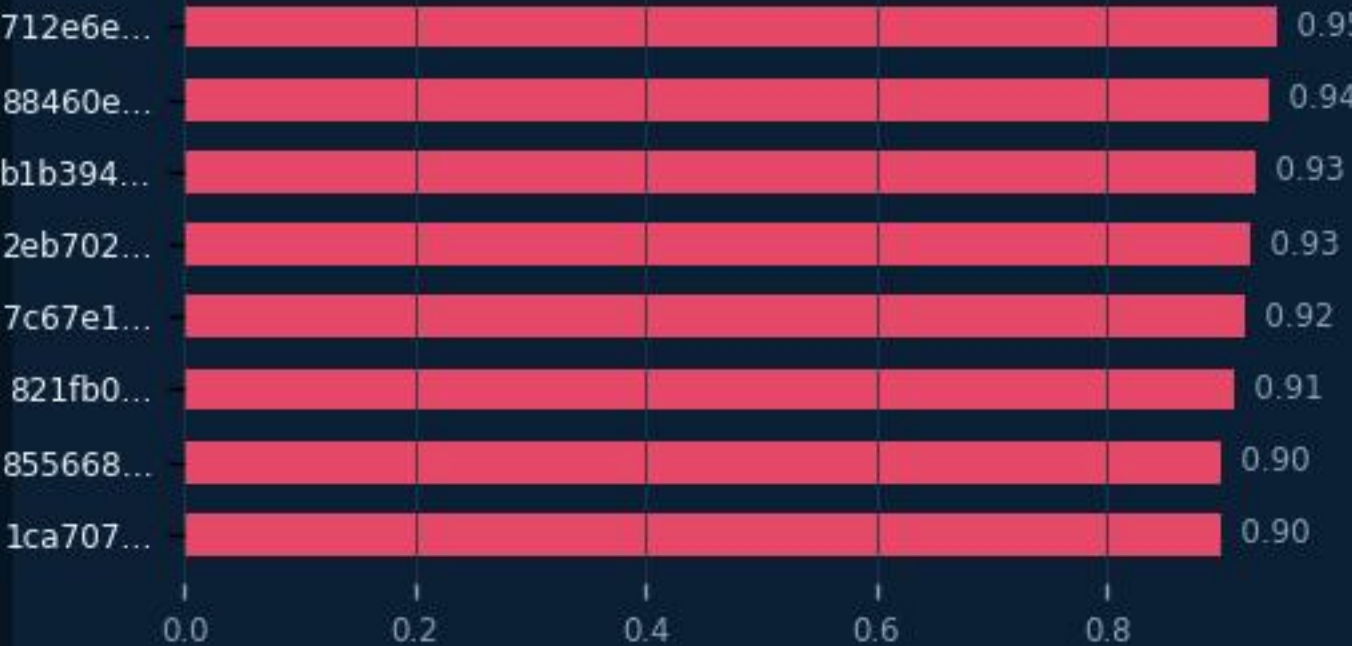
DRIVER WEIGHT

0.306

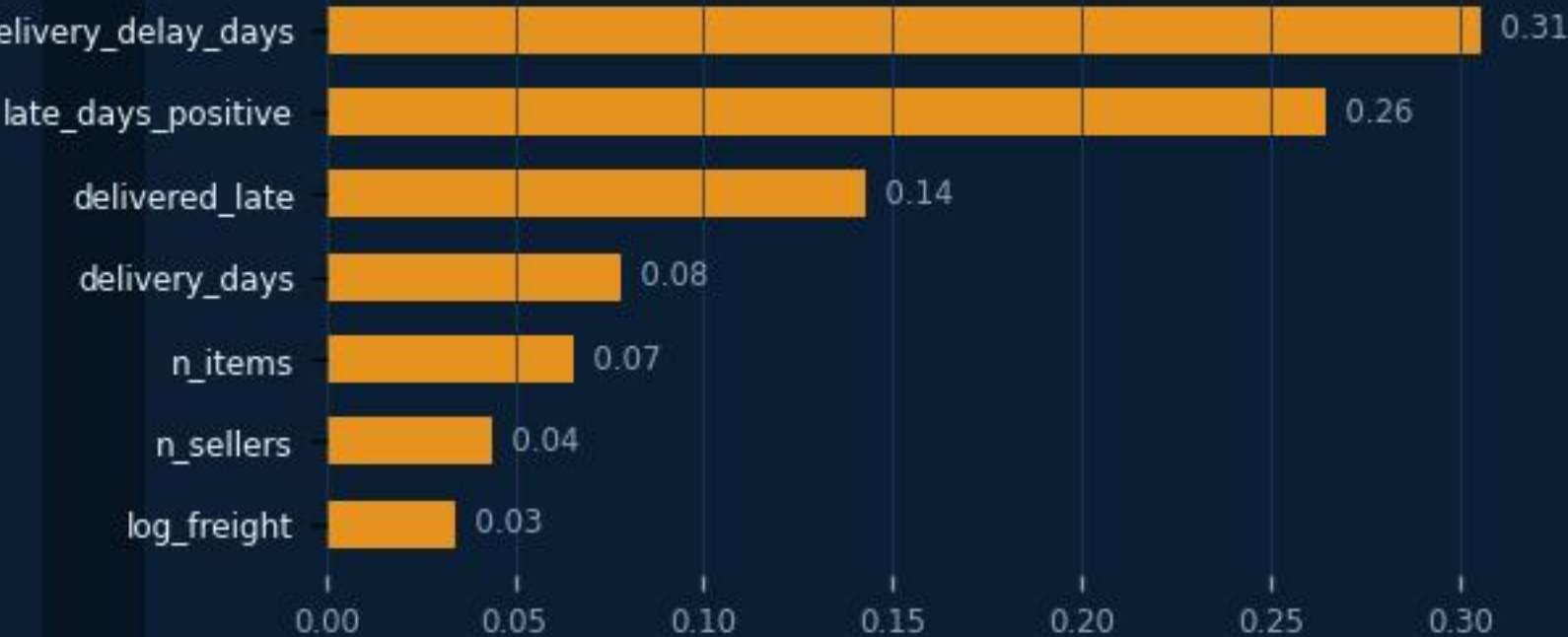
Revenue by Seller Segment



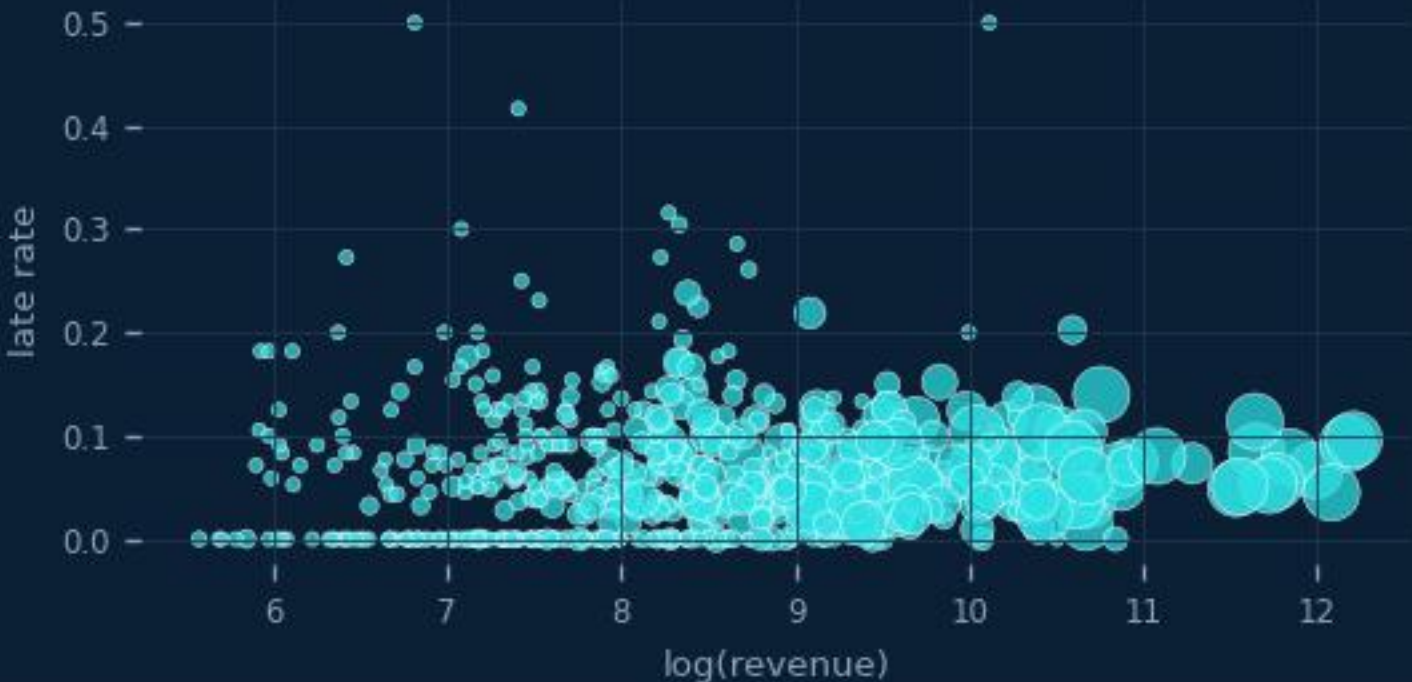
Seller Risk Ranking



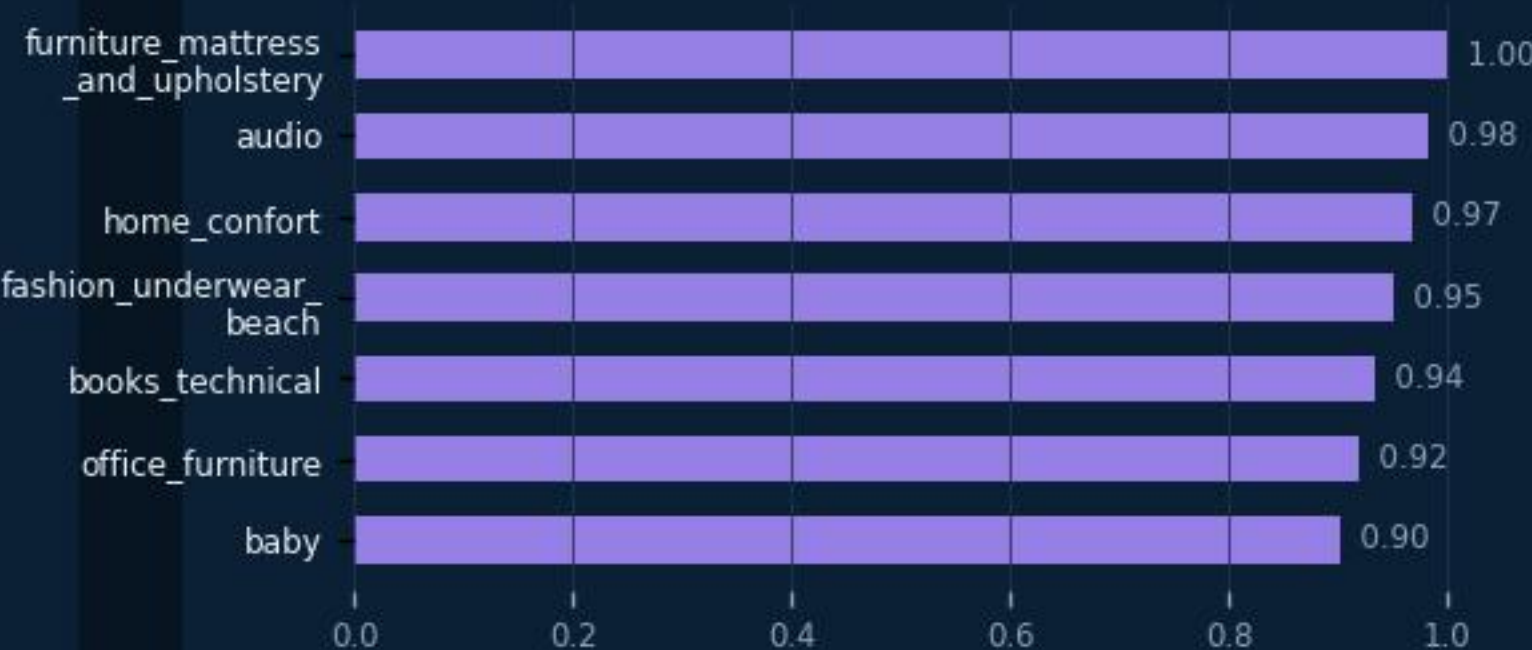
Review Risk Feature Importance



Seller Revenue vs Late Rate



Category Risk Typology



Seller Intelligence Reading

Core message

Seller analytics identifies risk concentration: the highest intervention target is seller 712e6ed8.... Review risk is mostly driven by delivery-delay variables, not only product economics.

BI + analytics layer demonstrated: seller segmentation, quality analytics, risk scoring, operational prioritization.

PRODUCT & CATEGORY INTELLIGENCE

Dashboard 4 / Product Analytics

Category Revenue • Strategic Risk • Quality Signals • Portfolio Priorities

CATEGORIES

73

TOP CATEGORY

health_beauty

TOP REVENUE

\$1.26M

STRATEGIC RISK

16

AVG REVIEW

4.11

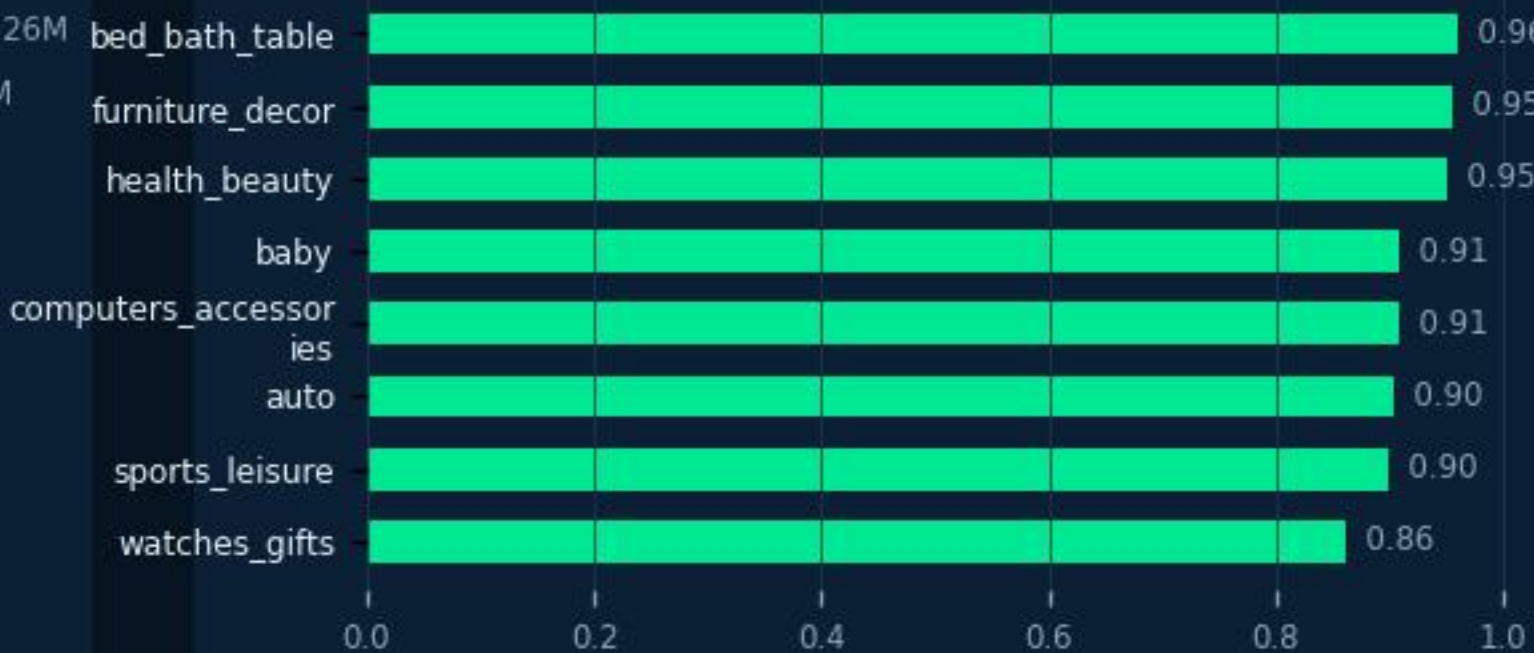
HIGHEST ROI CAT.

bed_bath_table

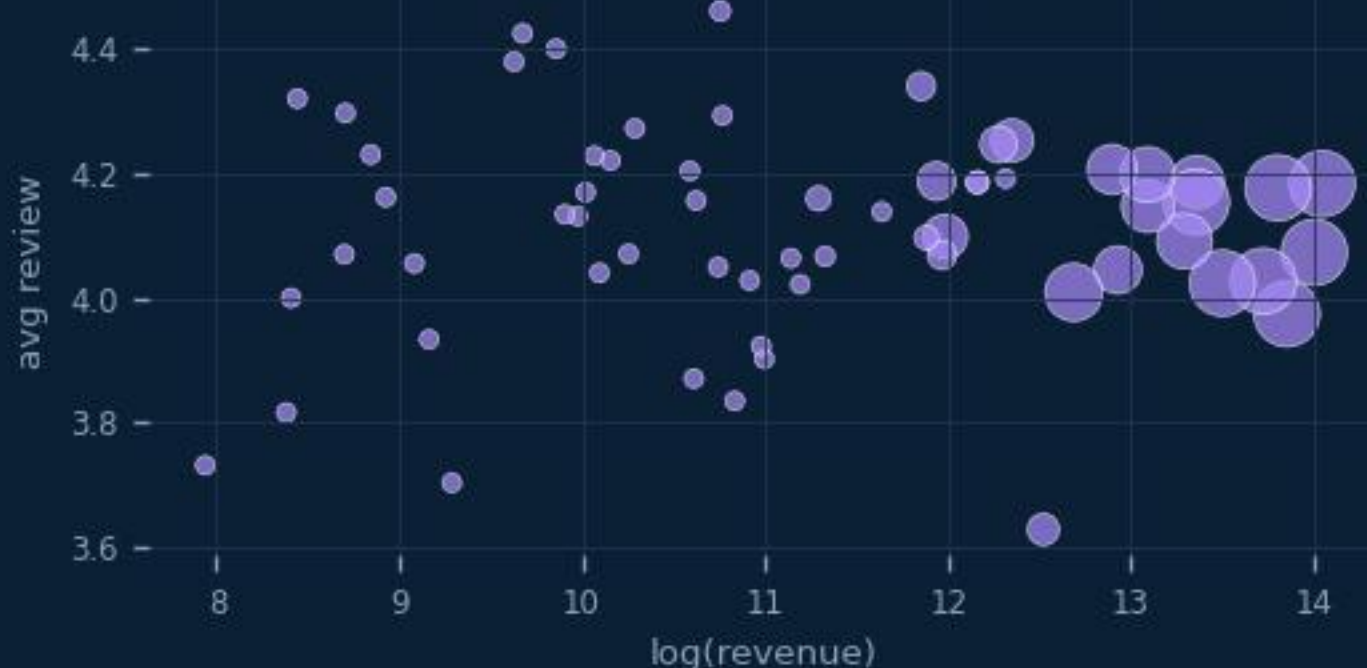
Top Categories by Revenue



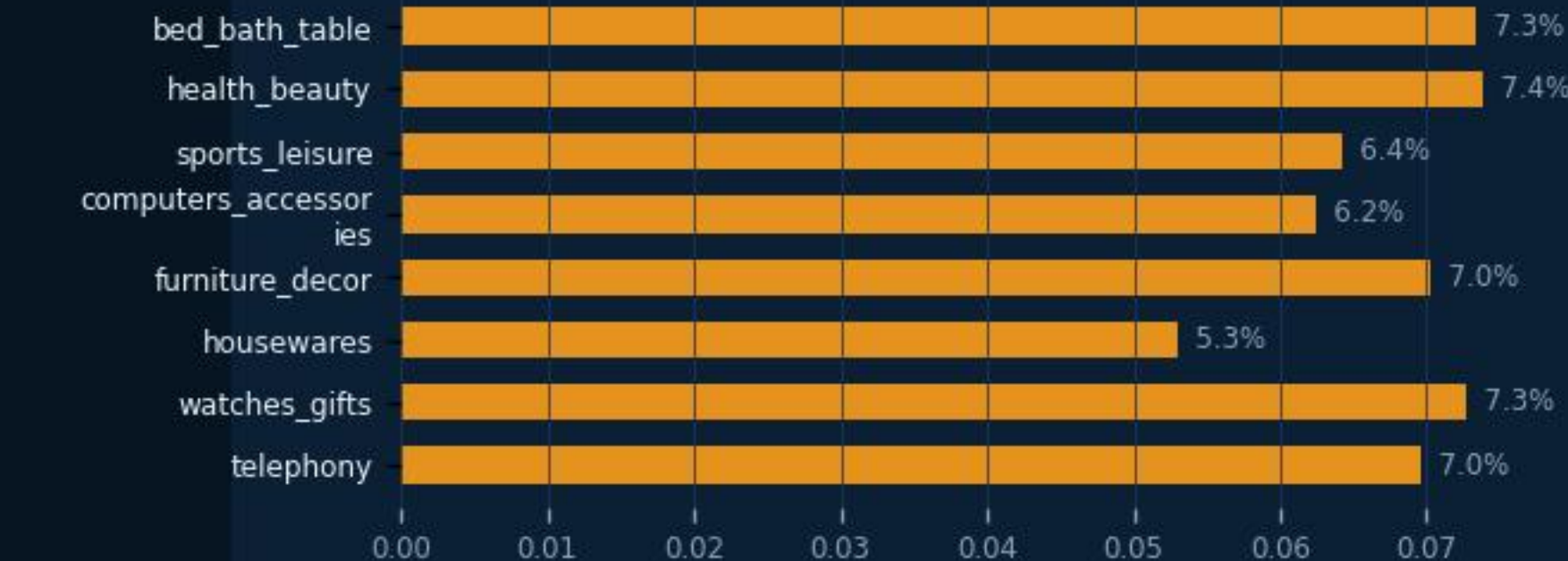
Category ROI Ranking



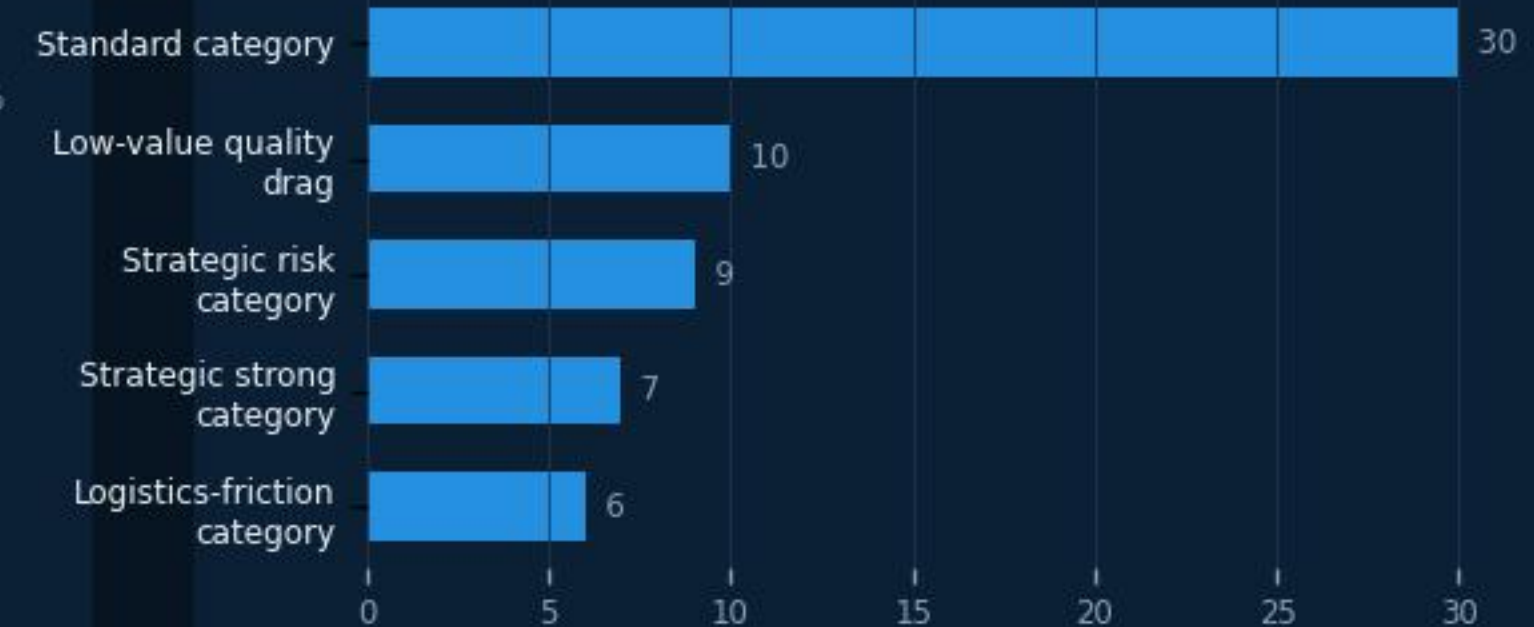
Quality vs Revenue Portfolio



Late Rate by Top Category



Category Types



Category Decision Reading

Core message

bed_bath_table has the highest category ROI score (0.96). Category intelligence combines revenue leverage, review performance, late-rate risk and network importance.

Power BI layer demonstrated: category drill-down, portfolio view, product performance reporting, opportunity ranking.

NETWORK SCIENCE & MARKETPLACE FRAGILITY

Dashboard 5 / Complex Systems

Centrality • Critical Nodes • Dependency Structure • Revenue at Risk

NETWORK NODES

3,108

CRITICAL SELLERS

25

CRITICAL CATEGORIES

25

TOP 1% REVENUE LOST

25.7%

TOP CATEGORY

health_beauty

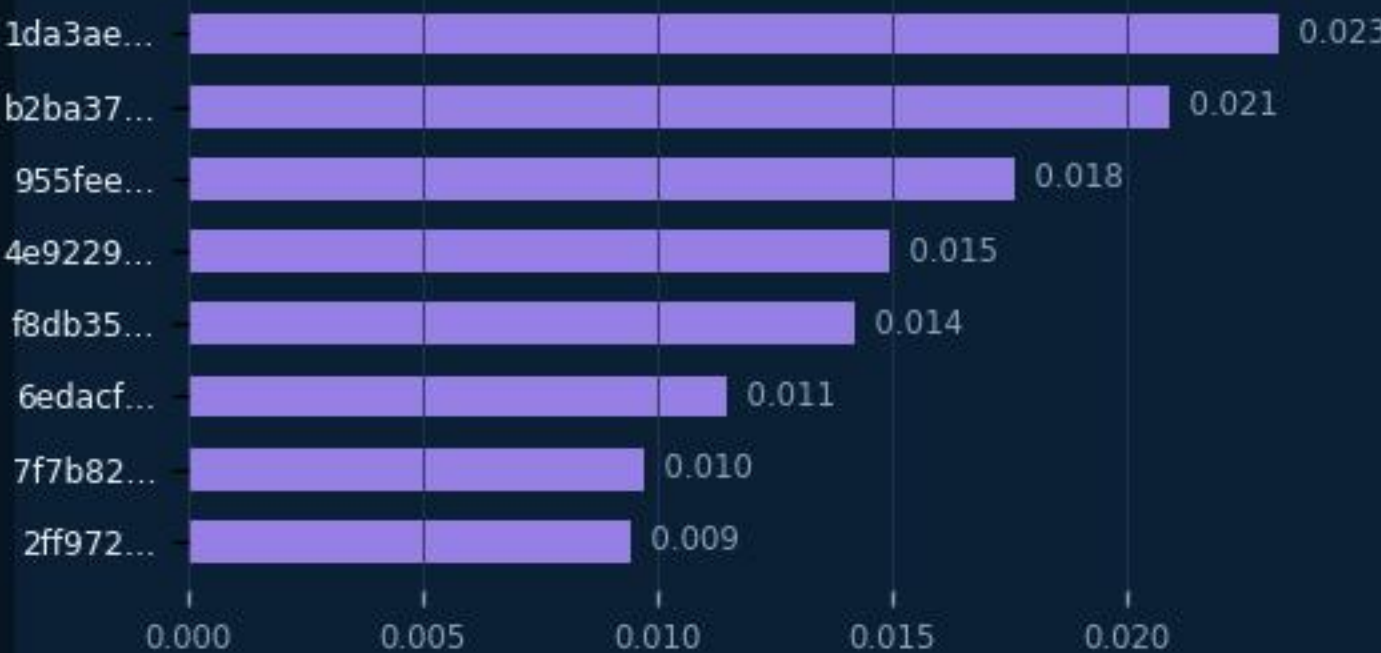
TOP SELLER NODE

1da3aeb7...

Critical Categories by Betweenness



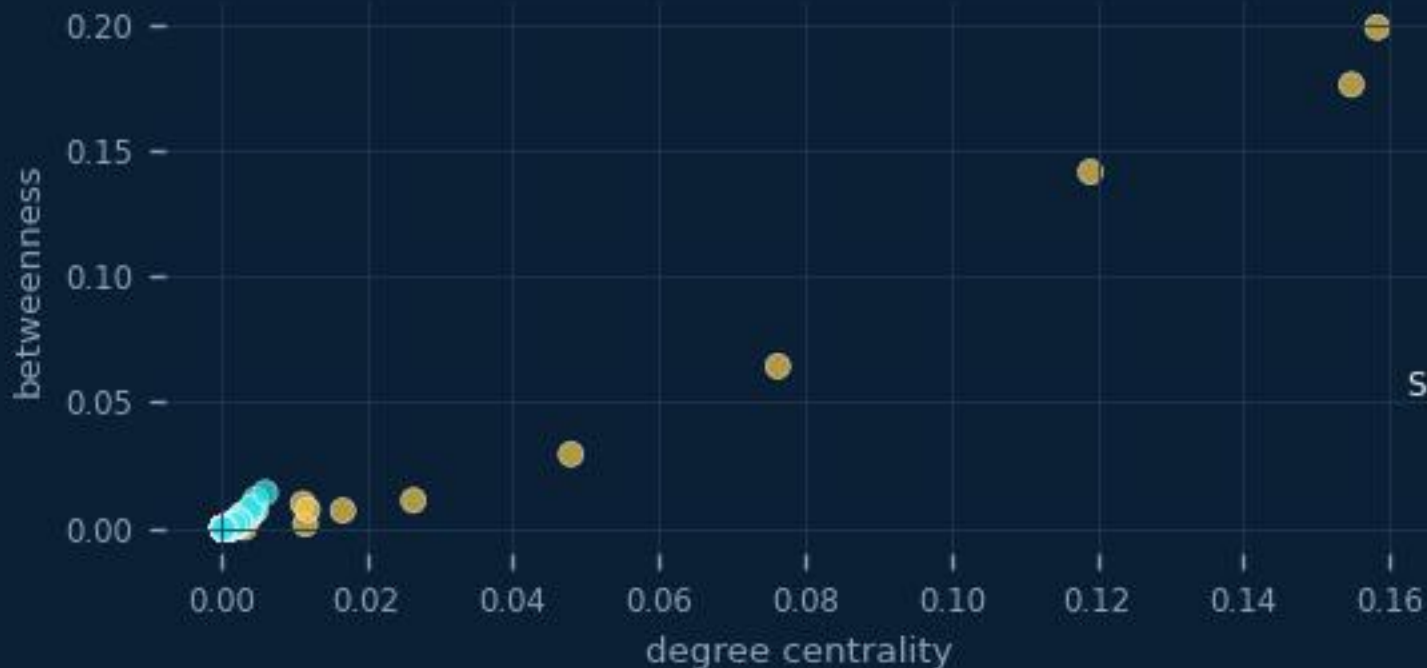
Critical Sellers by Betweenness



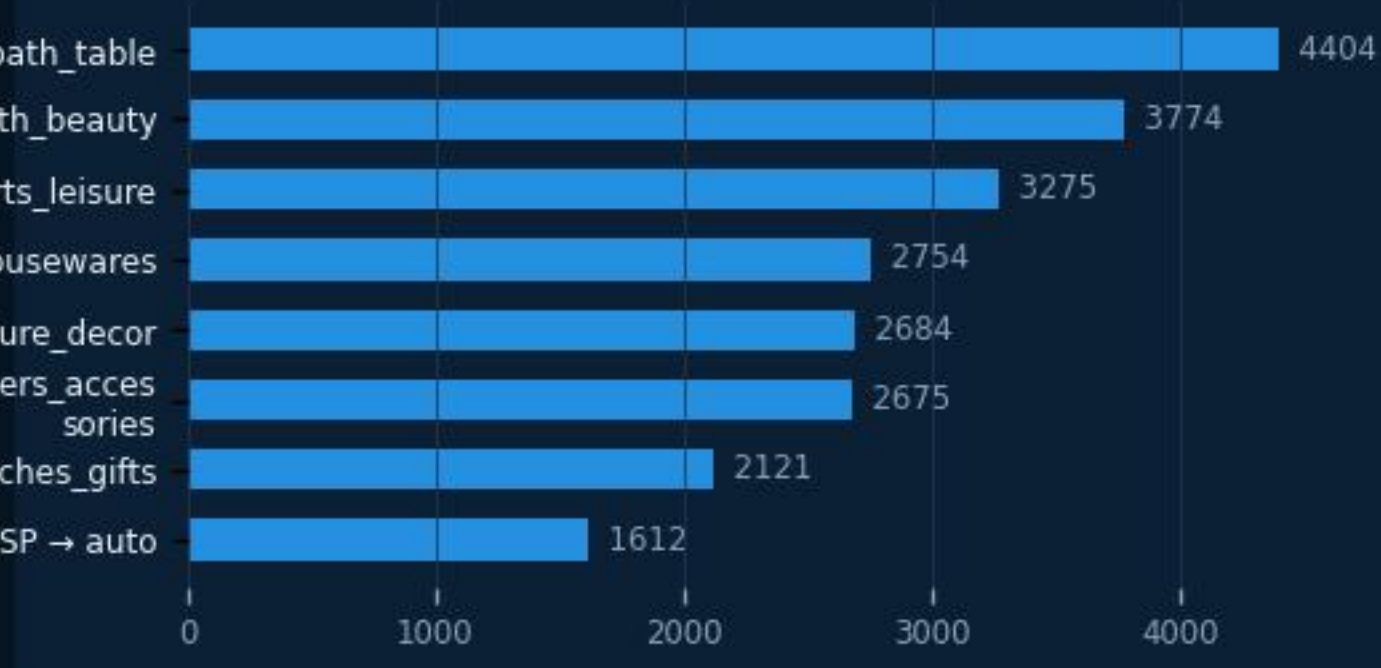
Fragility Scenarios



Network Centrality Cloud



State-Category Flow Snapshot



Network Science Reading

Core message

The marketplace is structurally fragile: removing the top 1% of sellers implies 25.7% revenue exposure. This moves analysis beyond dashboards into dependency and systemic-risk logic.

Advanced layer demonstrated: network centrality, fragility analysis, critical dependency detection, complex systems analytics.

DECISION INTELLIGENCE ENGINE

Intervention ROI • Prioritization • Revenue Impact • Recommended Actions

Dashboard 6 / Intervention Optimization

TOP INTERVENTION

Seller

TOP SCORE

0.991

BEST SELLER ROI

0.991

BEST CAT ROI

0.961

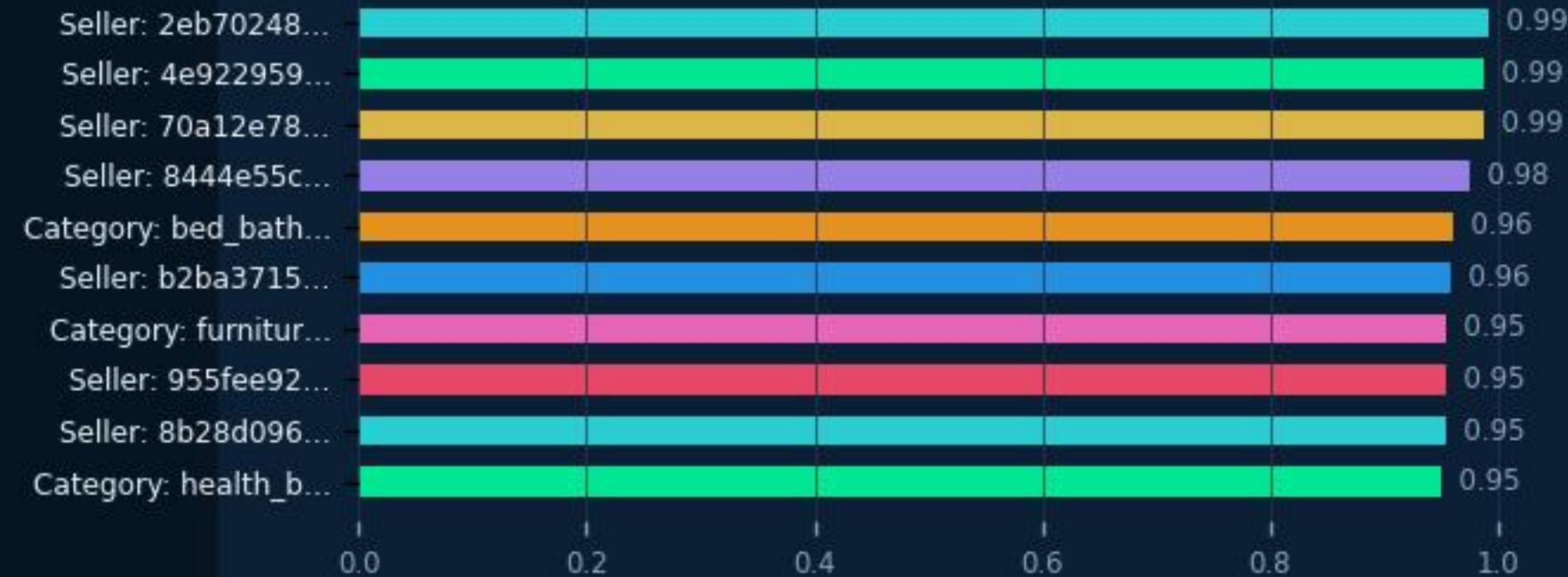
BEST CUST. SEG.

High-value one-t

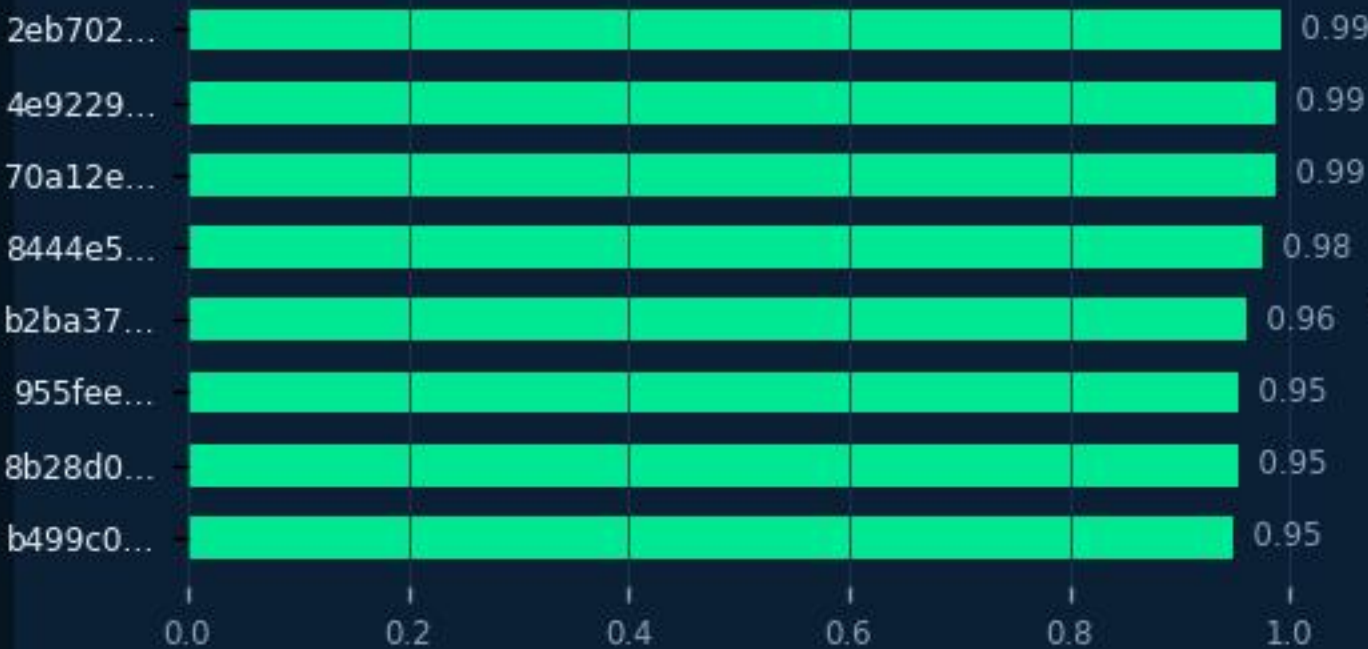
DELAY -50% GAIN

+0.32 review

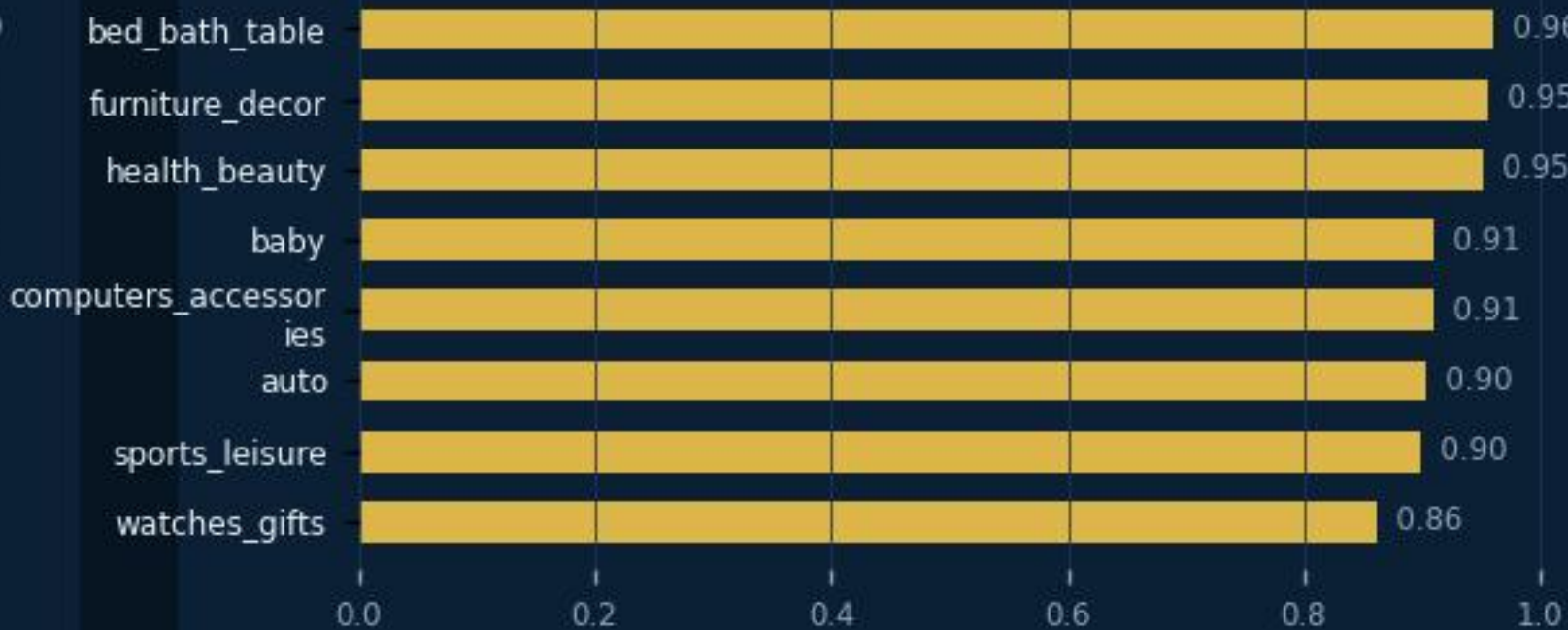
Marketplace Intervention Ranking



Seller ROI Ranking



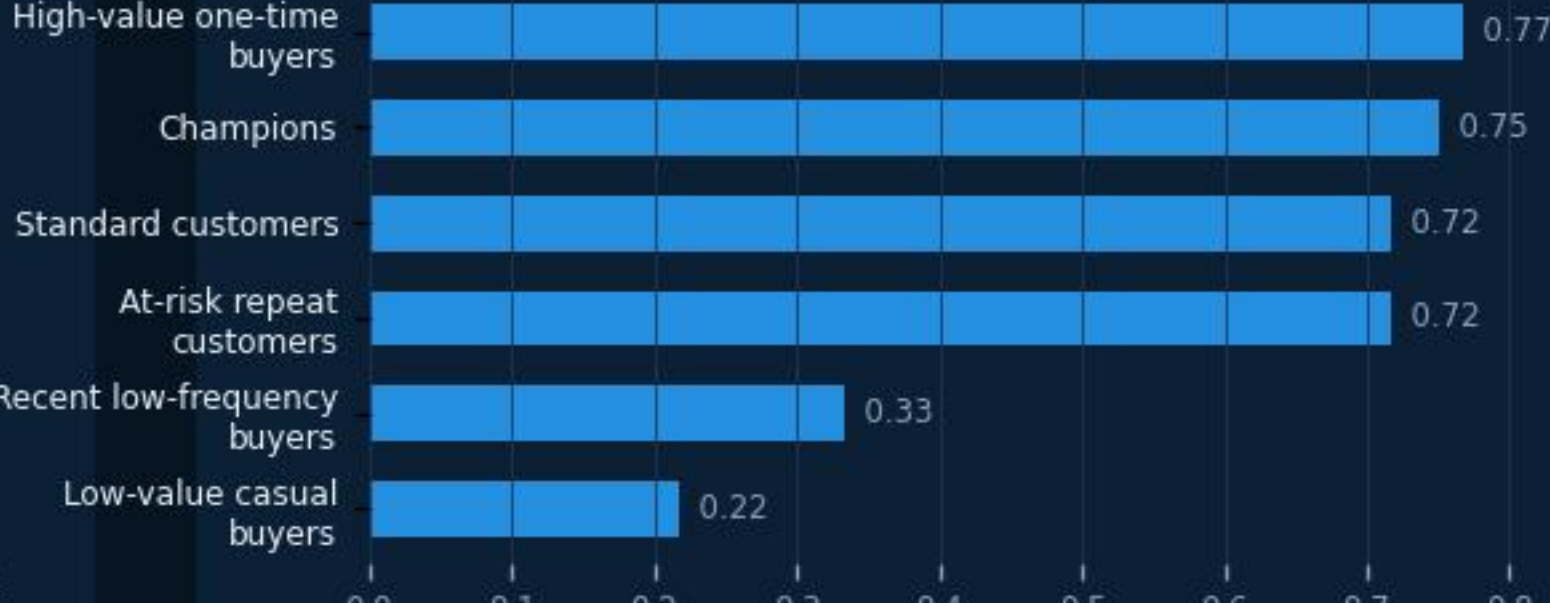
Category ROI Ranking



Logistics Improvement Scenarios



Customer Segment ROI



Decision Logic

Core message

The decision layer ranks sellers, categories and customer segments by expected intervention value. Priority begins with Seller 2eb70248... and score 0.991.

Decision layer demonstrated: ROI ranking, intervention prioritization, what-if scenarios, recommendation engine.